SOLUTION ARCHITECTURE

Project Title

Rising Waters

Project Subtitle

AI-Powered Flood Prediction System using Machine Learning

Introduction

The Solution Architecture of the Rising Waters project describes the overall structure of the application. It explains how different software components interact with each other to predict flood risk using Machine Learning.

Architecture Overview

The Rising Waters application follows a simple architecture consisting of the User Interface, Data Processing Layer, Machine Learning Model, and Prediction Result Module.

Architecture Components

1. User Interface

The user interacts with the application through a Streamlit web dashboard where rainfall and weather parameters are entered.

2. Input Layer

The application receives environmental parameters including:

- Annual Rainfall
- Jan–Feb Rainfall
- Mar–May Rainfall
- Jun–Sep Rainfall
- Oct–Dec Rainfall
- Temperature
- Humidity
- Cloud Cover

3. Data Processing Layer

The system validates the input values, cleans the data, and prepares it for prediction.

4. Machine Learning Layer

The trained XGBoost Machine Learning model processes the input values and predicts the flood risk.

5. Prediction Layer

The model generates the prediction result as High Flood Risk or Low Flood Risk.

6. Output Layer

The prediction is displayed on the Streamlit dashboard along with the flood risk status.

Technology Used

• Python
• Streamlit
• Pandas
• NumPy
• Scikit-learn
• XGBoost
• Joblib

Architecture Flow

User
│
▼
Enter Weather & Rainfall Data
│
▼
Input Validation
│
▼
Data Preprocessing
│
▼
Machine Learning Model (XGBoost)
│
▼
Flood Risk Prediction
│
▼
Display Result on Dashboard

Advantages

• Simple architecture
• Easy maintenance
• Fast prediction
• High accuracy
• Scalable design
• User-friendly interface

Conclusion

The Rising Waters architecture provides a reliable and efficient framework for predicting flood risk using Machine Learning. The modular design ensures better performance, maintainability, and future scalability.